Stochastic Storm Surge Modeling
using ADCIRC+SWAN through Parallel Computing

SCL Presentation
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J Franco Ray
Scientific Computing Laboratory
Department of Computer Science
University of the Philippines Diliman
jbrayo@up.edu.ph

Outline

Introduction

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- Storm surge predictions from different agencies are not completely consistent
- No CFD software for stochastic storm surge
- We need better and more realistic model!

Research Objective

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- Investigate parameter sensitivity

Related Studies

Storm Surge Modeling

Can be done using deterministic models

- ADCIRC+SWAN
- Delft3D-FLOW+WAVE
- SLOSH
- JMA

Only ADCIRC+SWAN can compute in parallel

Can also use statistical analysis based from past events

ADCIRC+SWAN

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- Finite-element in time and finite-difference in time
- Uses shallow water equations and continuity equation to obtain water velocity and elevation
- Grid flexibility, efficiency and accuracy well documented

Sensitivity Analysis

Wetting and Drying Algorithm

- If the water total depth is larger than H_{min}, the node remains "wet" and included in the rest of calculations.
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$$U = \frac{g\left(\zeta_{i-1} - \zeta_i\right)}{\tau_i \Delta x_i}$$

- This criterion almost becomes a height restriction if the adj node's free surface circulation is sufficiently larger than its own.

Bottom Friction

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- Note that $\tau \propto \tau_0$
- τ_i increases in shallower, near-shore waters and directly affects the speed at which waves inundate or recede.

GWCE Weighing Factor

- The parameter G is the numerical parameter that controls the balance between the wave continuity equation and the primitive continuity equation.

$$\frac{\partial \zeta}{\partial t} + G\zeta - \nabla \cdot M = 0$$

- As $G \to 0$: pure wave equation, $G \to \infty$: pure primitive equation.

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- If G is set too high, solution has spurious oscillations that prevent the model from capturing behavior of waves.

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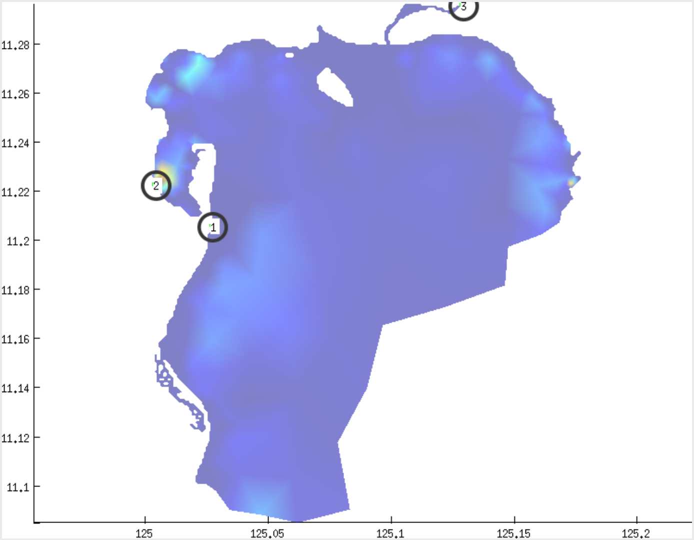
- Three dimensional analysis of parameters is very hard.

Methodology
observation points



fort.15 Input File

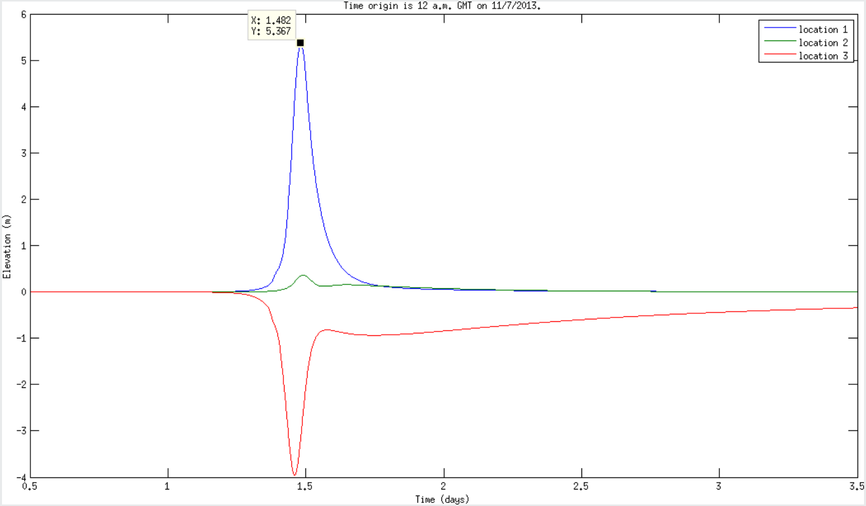
fort.15

```
SCL-ADCIRC-Haiyan-Typhoon-Track-Surge-Model
ADCIRC-V5-Feb1115
1 1E100 1 100 1E100                  ! NFOVER - NONFATAL ERROR OVERRIDE OPTION
-1                                   ! NABOUT - ABREVIATED OUTPUT OPTION PARAMETER
1                                    ! NSCREEN - UNIT 6 OUTPUT OPTION PARAMETER
0                                    ! IHOT - HOT START PARAMETER
2                                    ! ICS - COORDINATE SYSTEM SELECTION PARAMETER
0                                    ! IM - MODEL SELECTION PARAMETER
0                                    ! NOLIBF - BOTTOM FRICTION TERM SELECTION PARAMETER
0                                    ! NOLIFA - FINITE AMPLITUDE TERM SELECTION PARAMETER
0                                    ! NOLICA - SPATIAL DERIVATIVE CONVECTIVE TERM SELECTION PARAMETER
0                                    ! NOLICAT - TIME DERIVATIVE CONVECTIVE TERM SELECTION PARAMETER
0                                    ! NWP - VARIABLE BOTTOM FRICTION AND LATERAL VISCOSITY OPTION PARAMETER
1                                    ! NCOR - VARIABLE CORIOLIS IN SPACE OPTION PARAMETER
1                                    ! NTIP - TIDAL POTENTIAL OPTION PARAMETER
8                                    ! NWS - WIND STRESS AND BAROMETRIC PRESSURE OPTION PARAMETER
0                                    ! NRAMP - RAMP FUNCTION OPTION
9.80665                              ! ACCELERATION DUE TO GRAVITY - DETERMINES UNITS
0.01                                 ! G - WEIGHTING FACTOR IN GWCE
60                                   ! DT - TIME STEP (IN SECONDS)
0.00                                 ! STATIM - STARTING TIME (IN DAYS)
0.00                                 ! REFTIM - REFERENCE TIME (IN DAYS)
2013 11 07 00 01 0.8 21600           ! YYYY MM DD HH24 StormNumber BLAdjustmentFactor RSTIMINC
3.0                                  ! RNDAY - TOTAL LENGTH OF SIMULATION (IN DAYS)
2.0                                  ! DRAMP - DURATION OF RAMP FUNCTION
0.35 0.30 0.35                       ! A00 B00 C00 - TIME WEIGHING FACTORS IN GWCE
0.10 0 0 0.10                        ! H_min - MIMINMUM CUTOFF DEPTH, U_min
125.027139 11.205806                 ! center lon and lat for CPP
0.15                                 ! TAU_0 - BOTTOM FRICTION COEFFICIENT
2.0                                  ! ESLM - LATERAL EDDY VISCOSITY COEFFICIENT; IGNORED IF NWP =1
3.0314E-5                            ! CORI - CORIOLIS PARAMETER - IGNORED IF NCOR = 1
8                                    ! NTIF - TOTAL NUMBER OF TIDAL POTENTIAL CONSTITUENTS BEING FORCED
K1
0.141565 0.00001160579658 0.690 0.91229 49.73
O1
```

work/fort.15 19:1 (1,94) CRLF UTF-8 Plain Text

Original Inputs

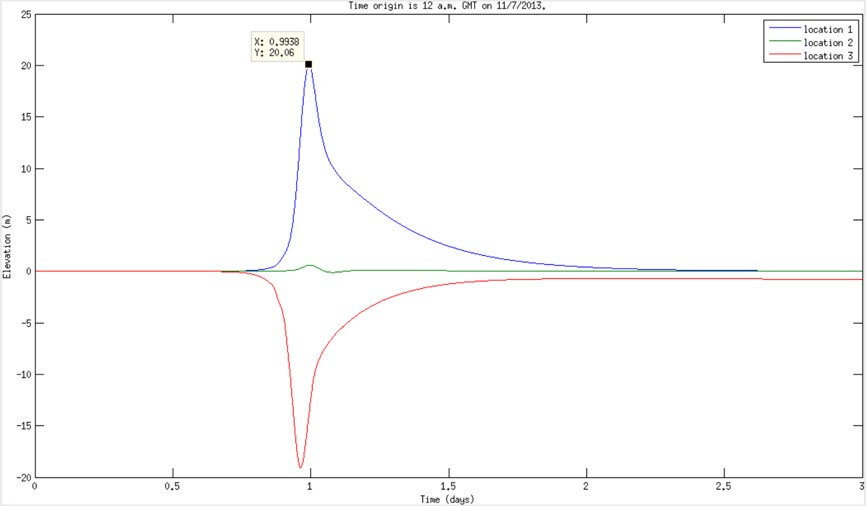
H_{min}=0.10, G=0.01, τ_0=0.15, $\hat{\zeta}$=5.367



Results

Water Cutoff Depth

H_{min} **0.01** 0.10 0.15 0.20 0.25 0.30 0.40 0.50



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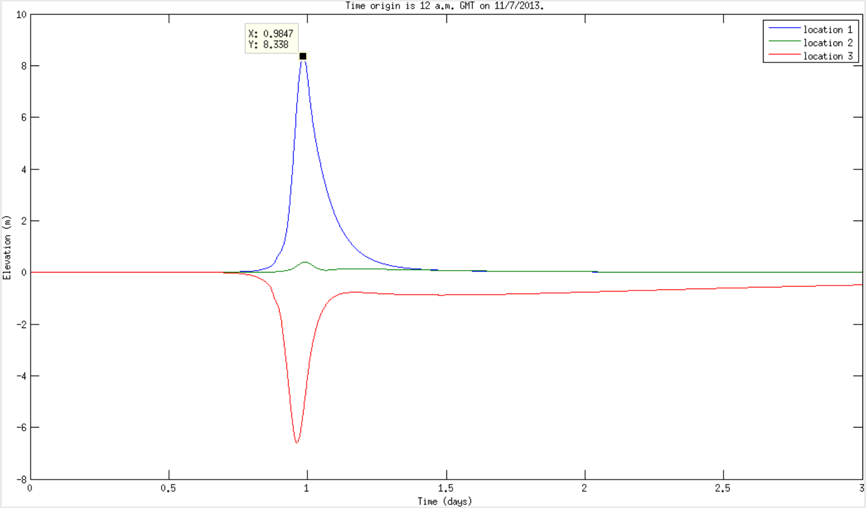

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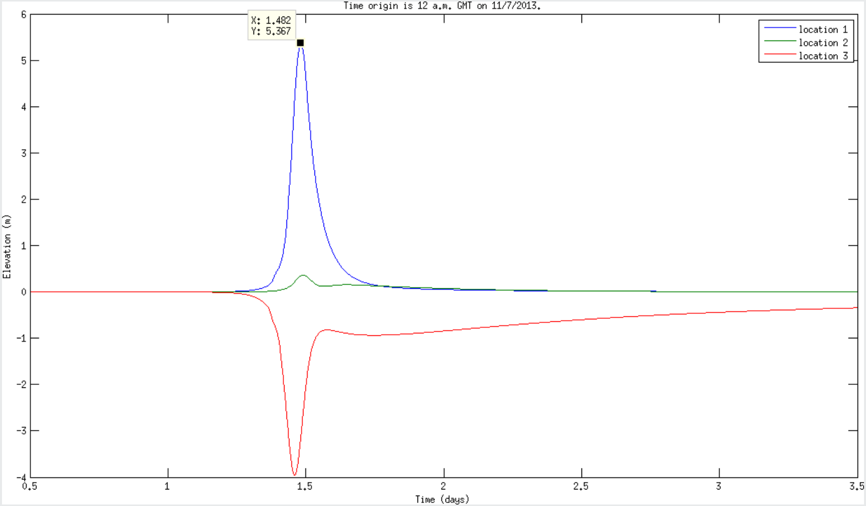

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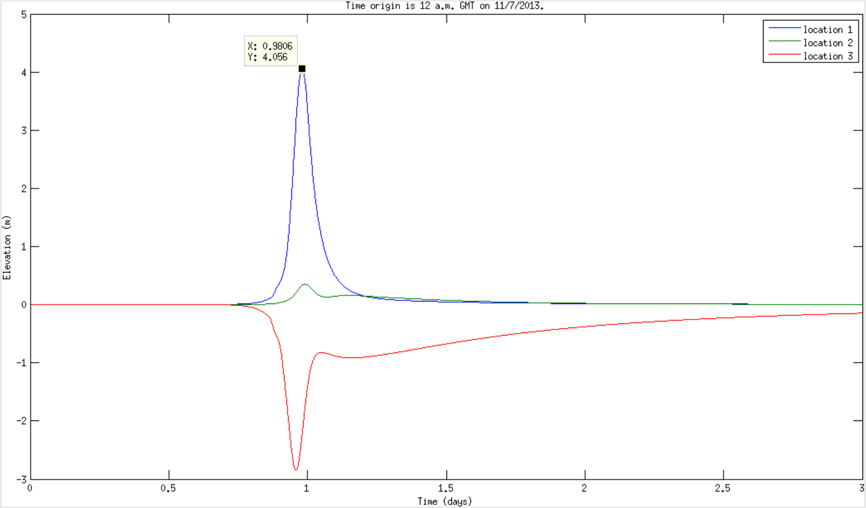

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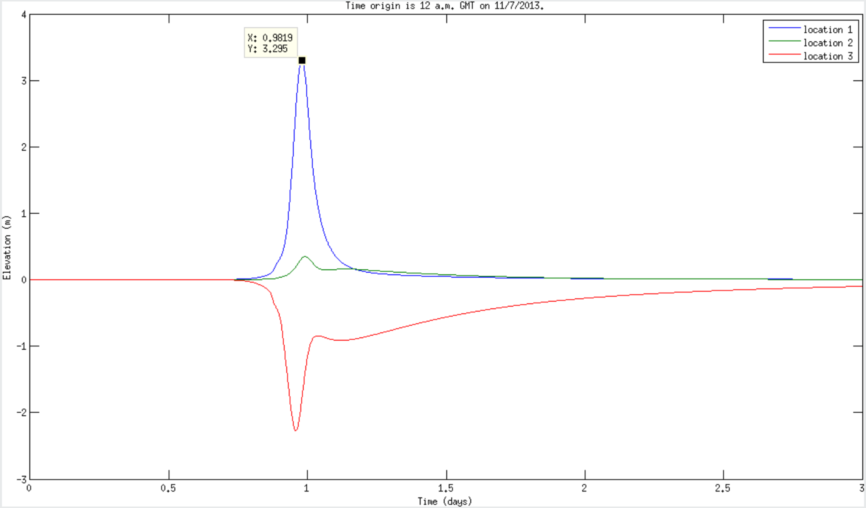

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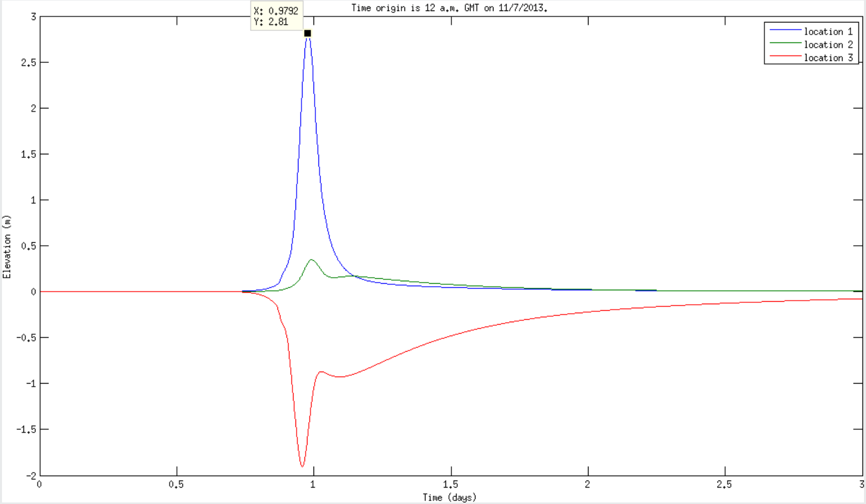

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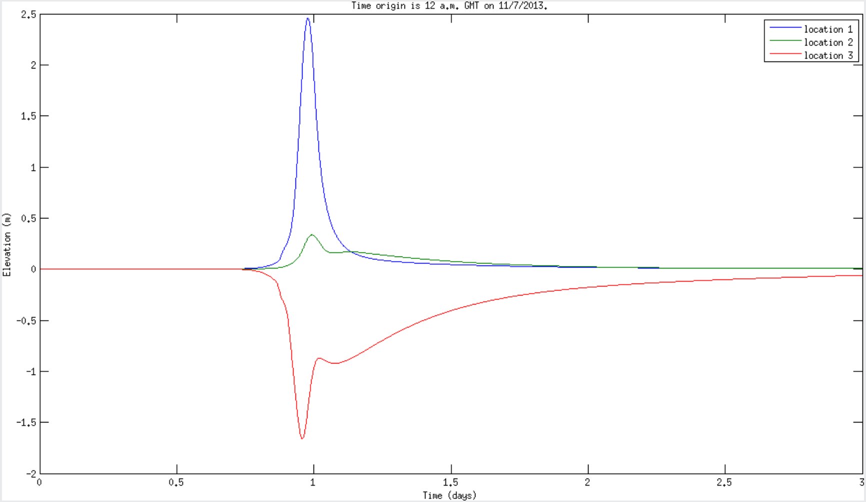

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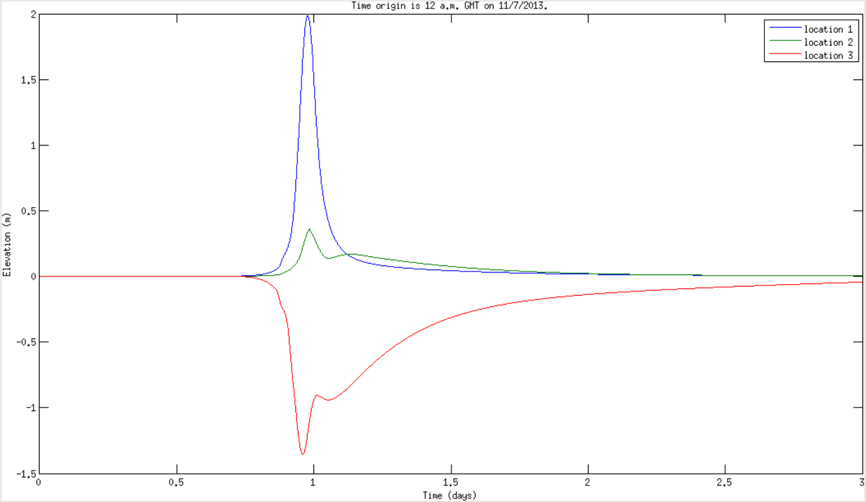

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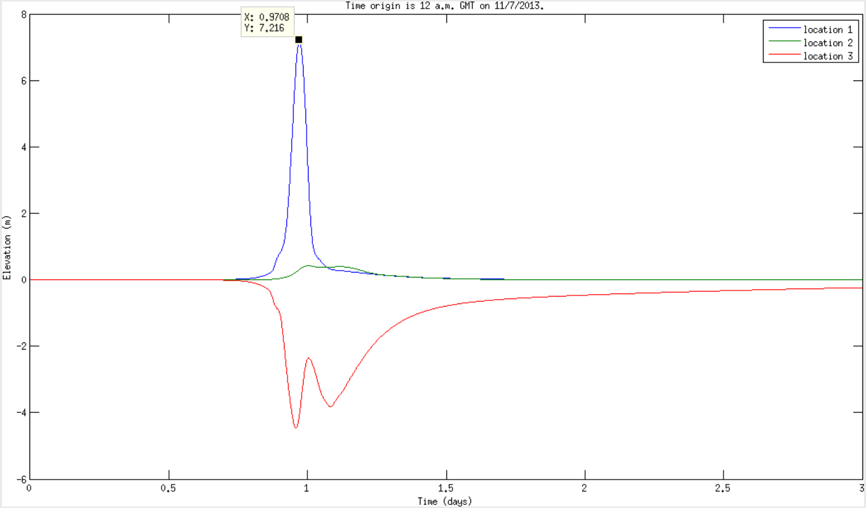

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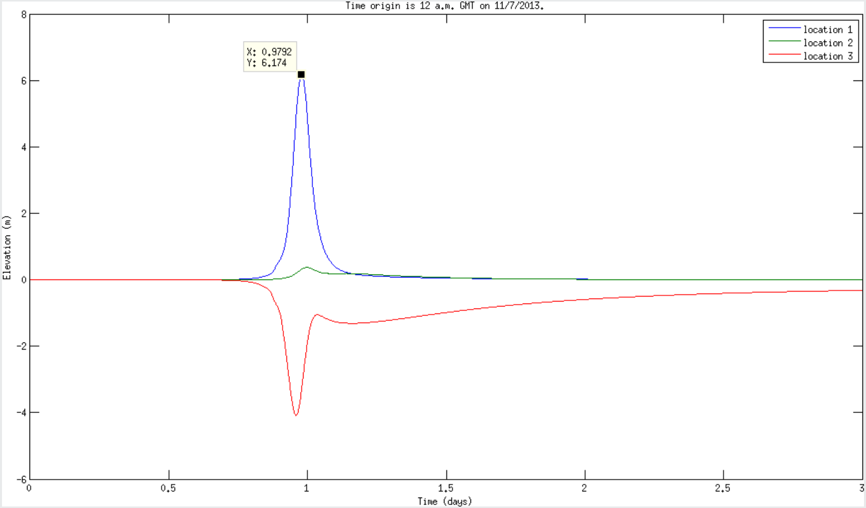

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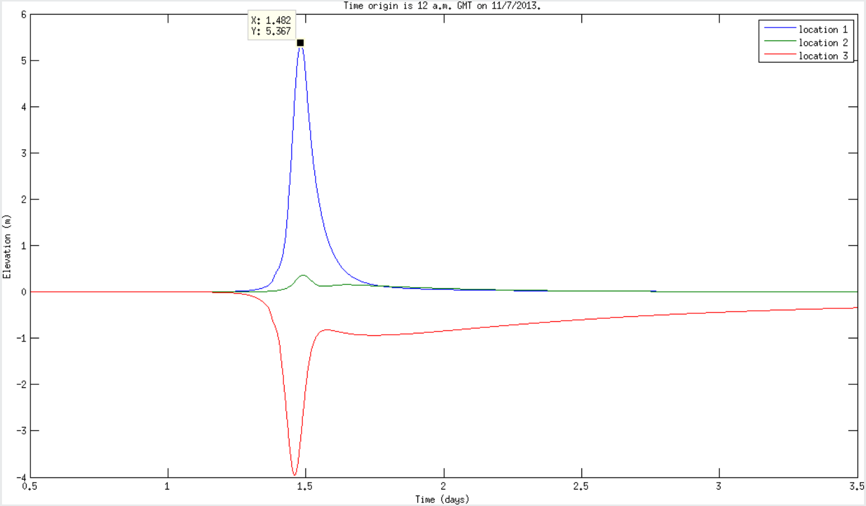

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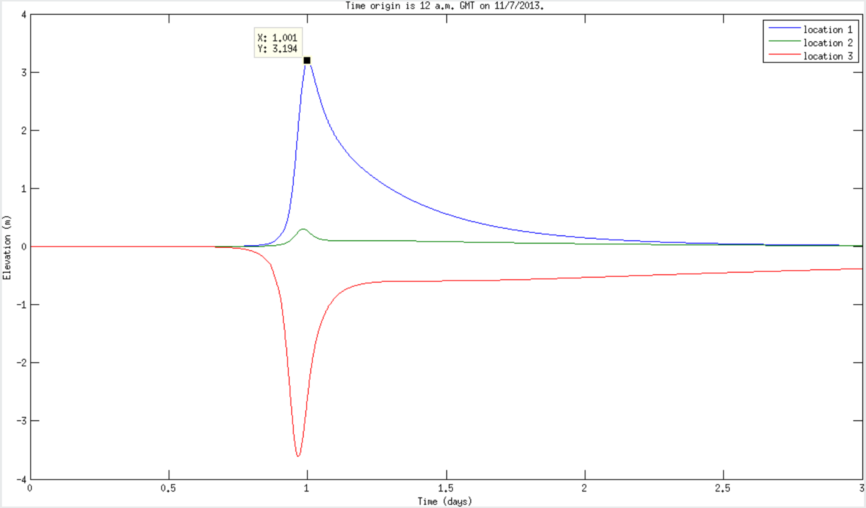

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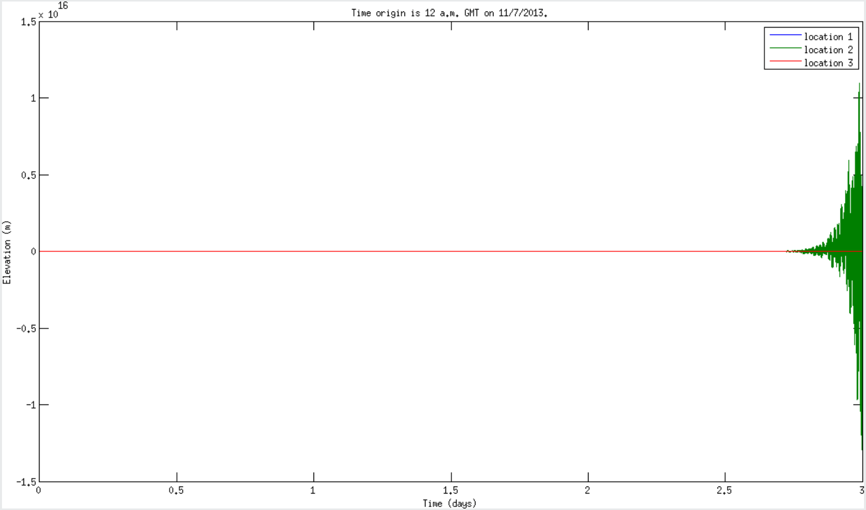
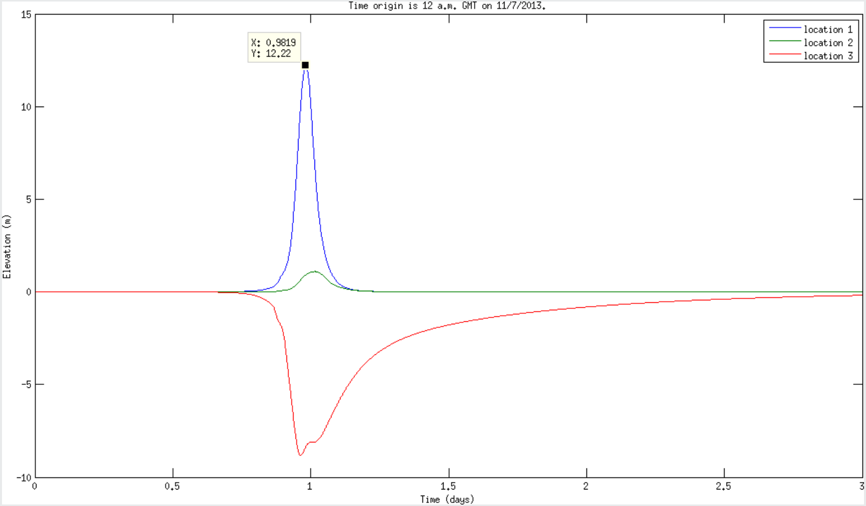


Bottom Friction

τ_0 **0.01** 0.05 0.10 0.15 0.20



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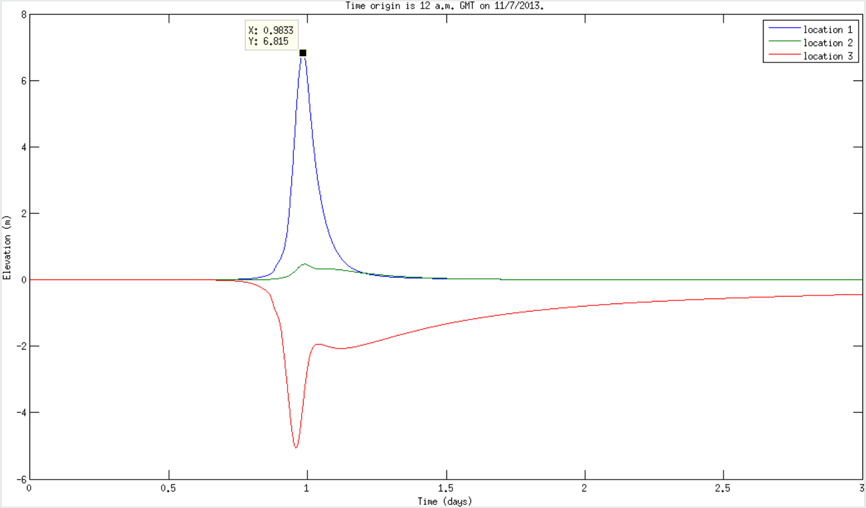

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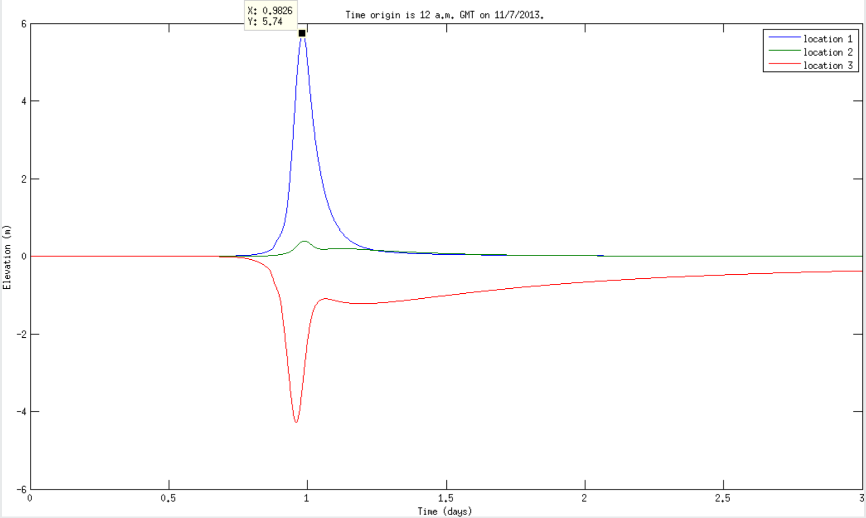

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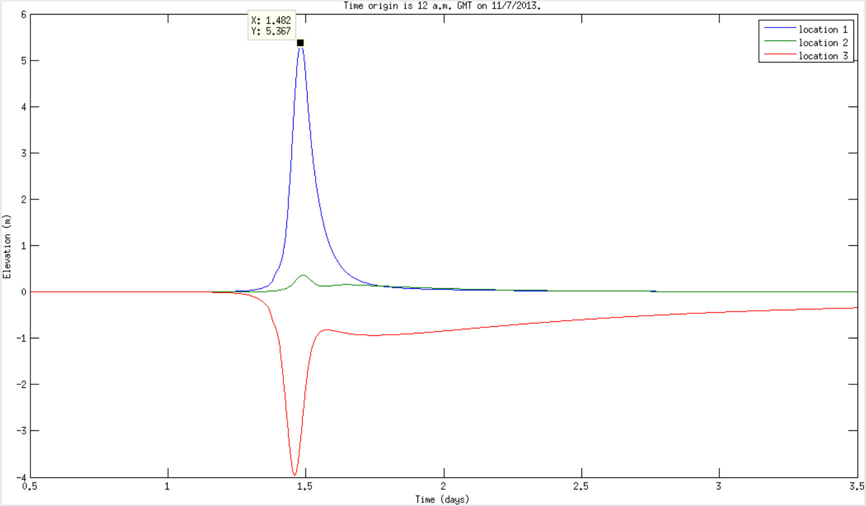

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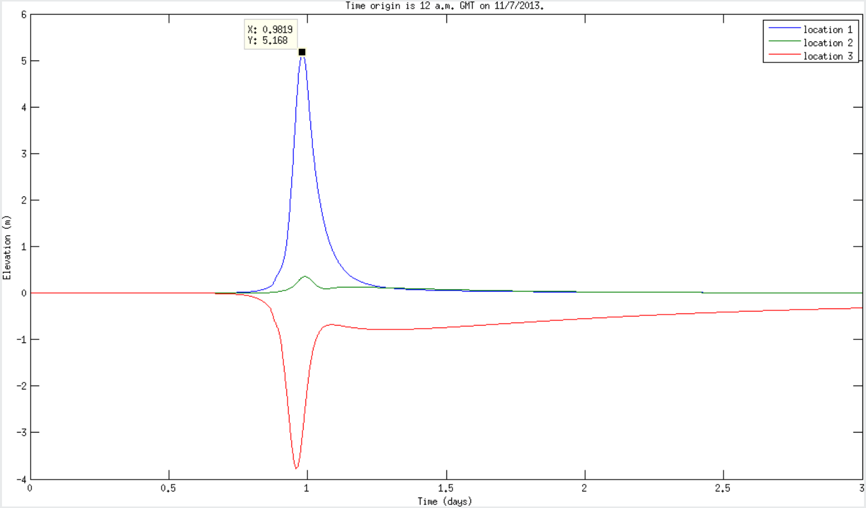
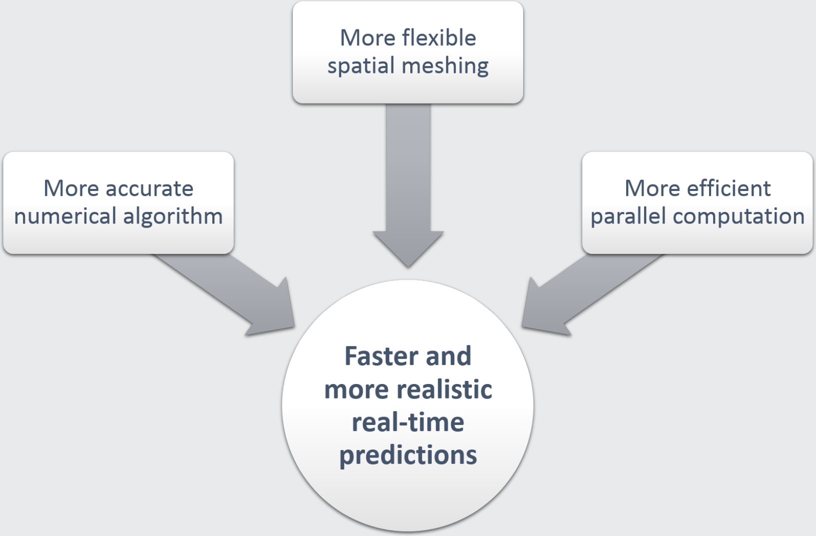


Observations

- Lower H_{min}, higher surge
- Higher G, more oscillations, unstable. Lower G, higher surge height; water subsides slower
- Lower τ_0, higher surge height; water subsides slower

Conclusion


More flexible spatial meshing
More accurate numerical algorithm
More efficient parallel computation
Faster and more realistic real-time predictions

Selected References

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Thank you for your attention!